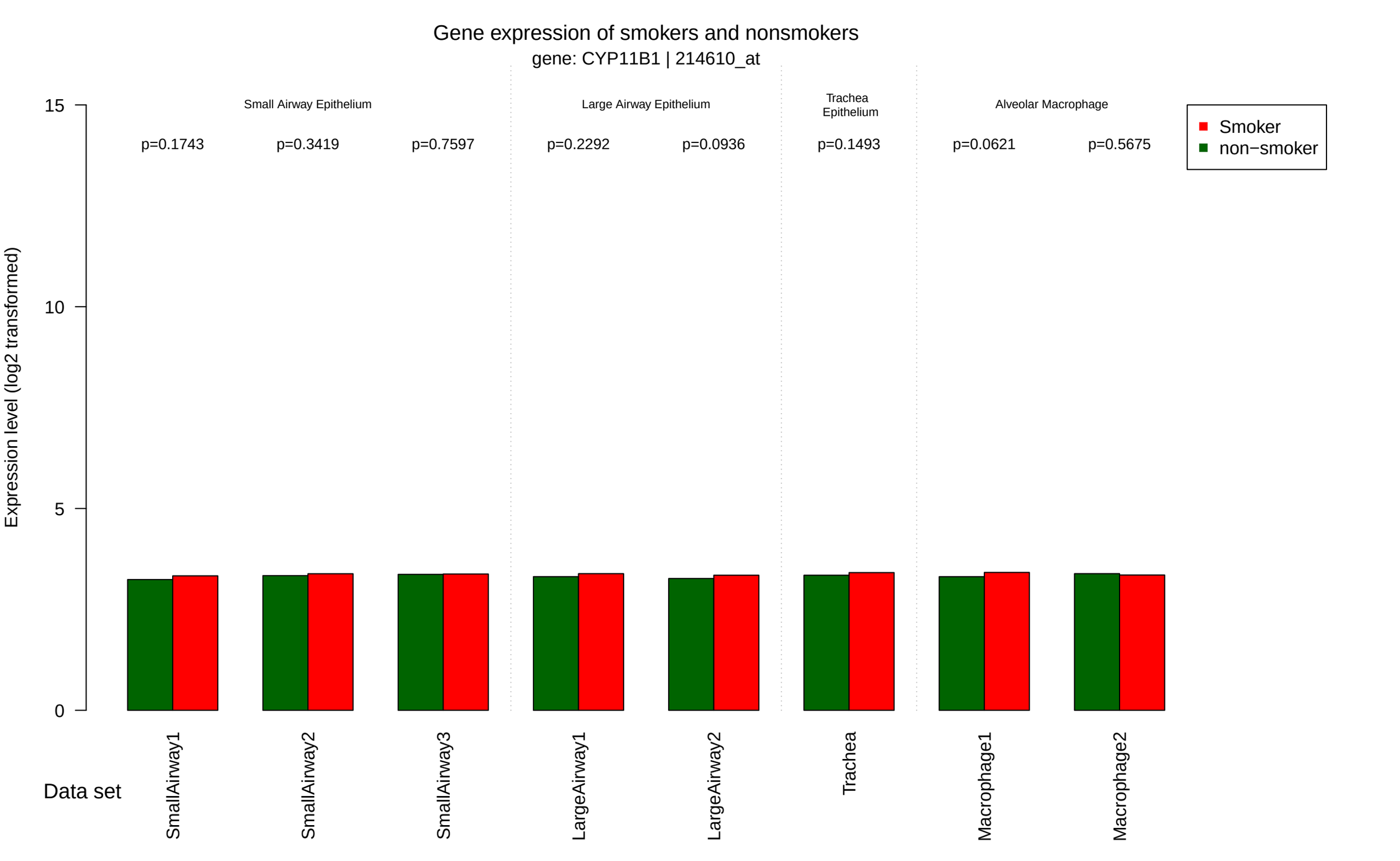
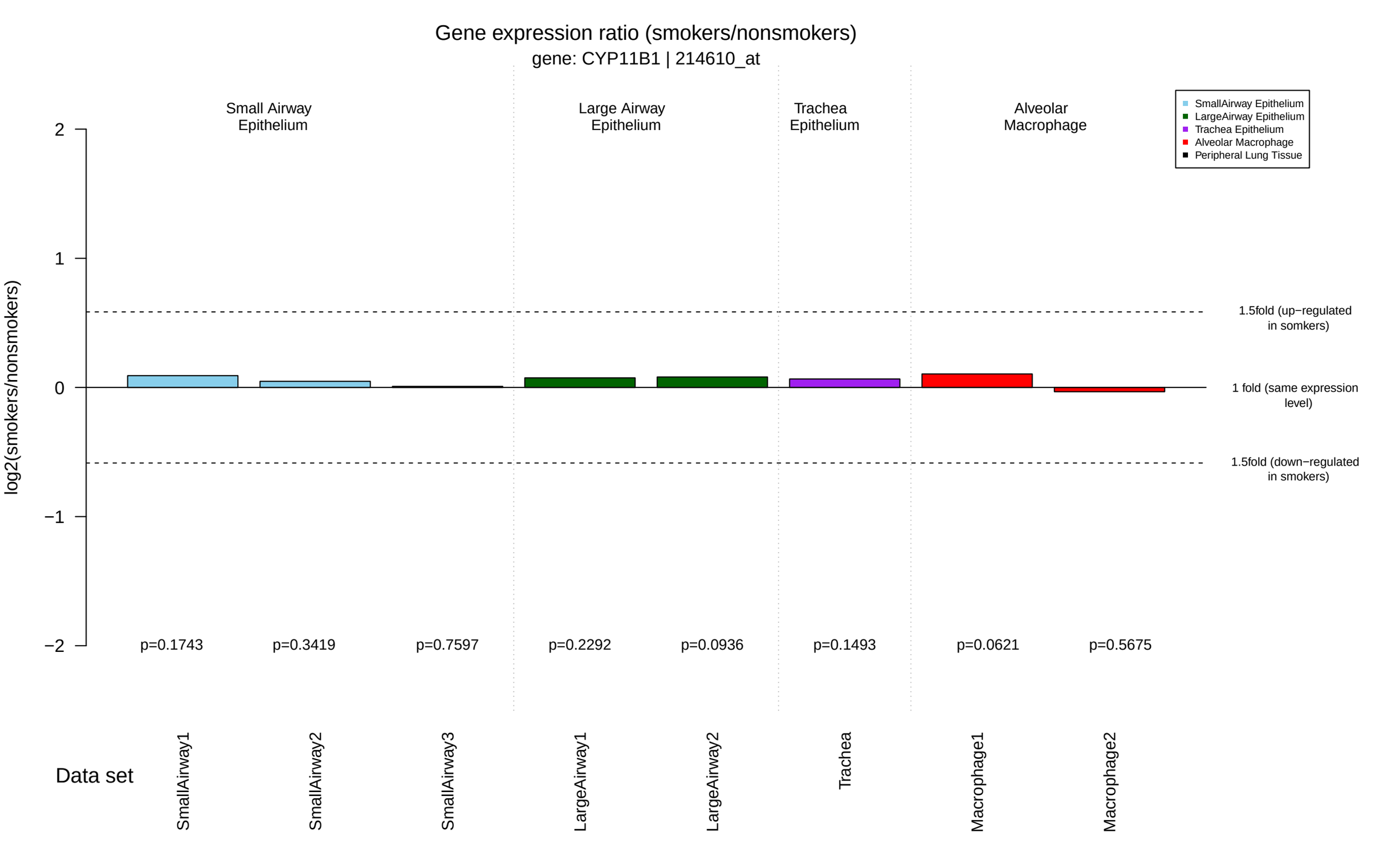
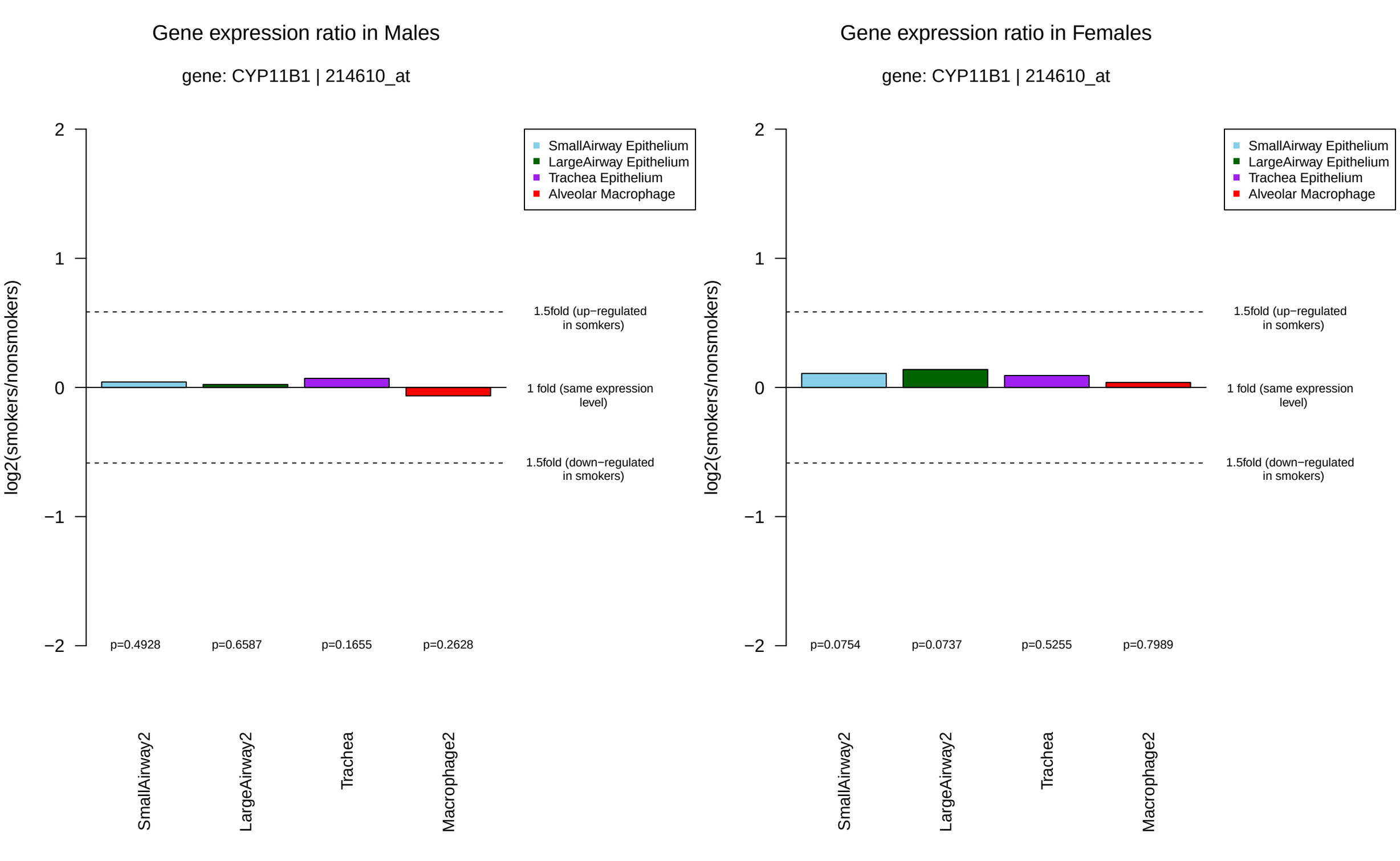
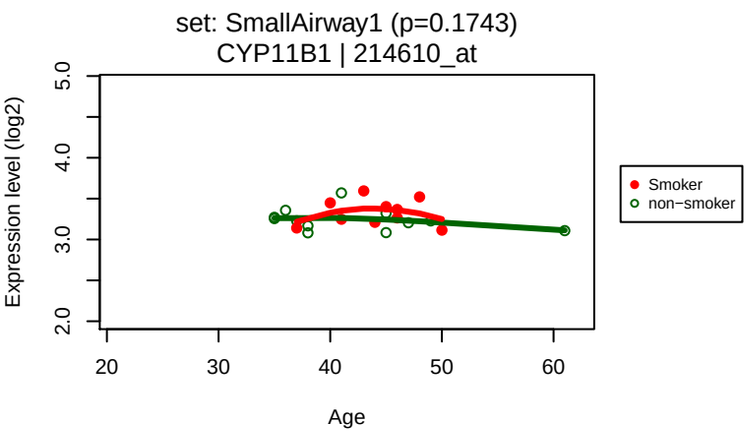
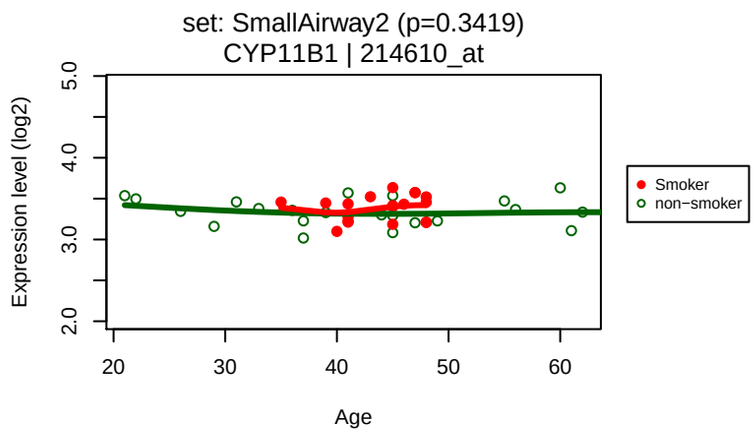
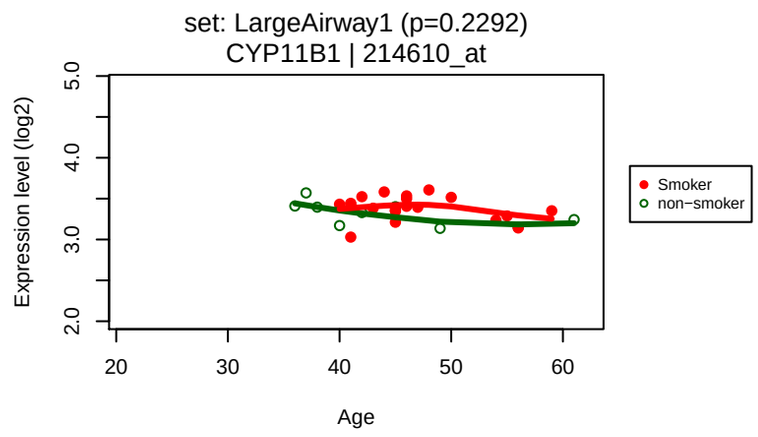

Gene expression of smokers and nonsmokers
gene: CYP11B1 | 214610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1743
p=0.3419
p=0.7597
p=0.2292
p=0.0936
p=0.1493
p=0.0621
p=0.5675
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

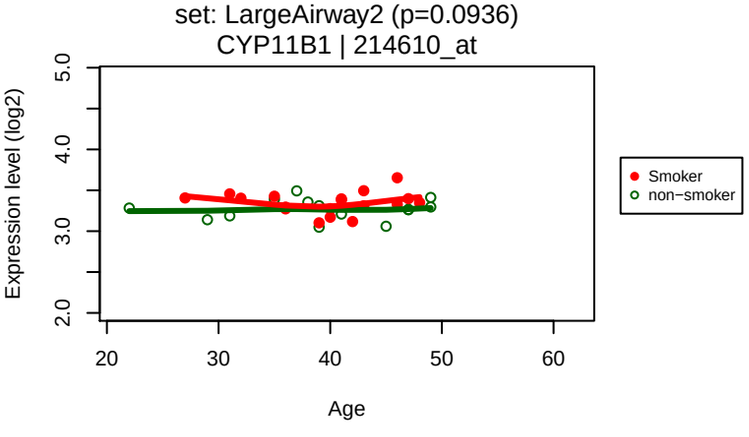

Gene expression ratio (smokers/nonsmokers)
gene: CYP11B1 | 214610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1743
p=0.3419
p=0.7597
p=0.2292
p=0.0936
p=0.1493
p=0.0621
p=0.5675
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

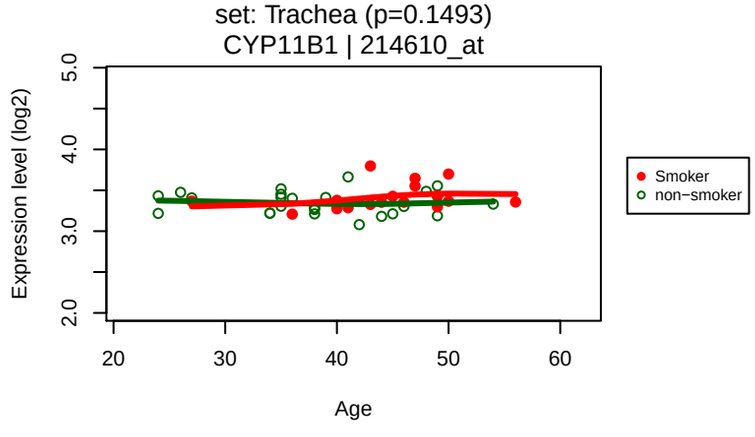

Gene expression ratio in Males
gene: CYP11B1 | 214610_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4928
p=0.6587
p=0.1655
p=0.2628
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP11B1 | 214610_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0754
p=0.0737
p=0.5255
p=0.7989
SmallAirway2
LargeAirway2
Trachea
Macrophage2

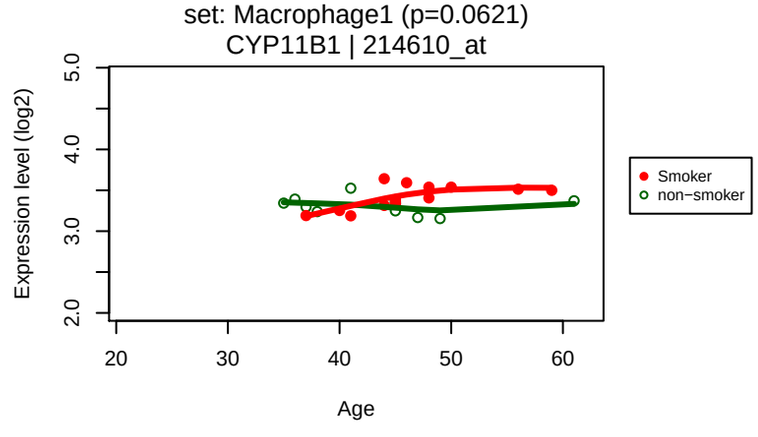

set: SmallAirway1 (p=0.1743)
CYP11B1 | 214610_at
Expression level (log2)
Age
Smoker
non−smoker

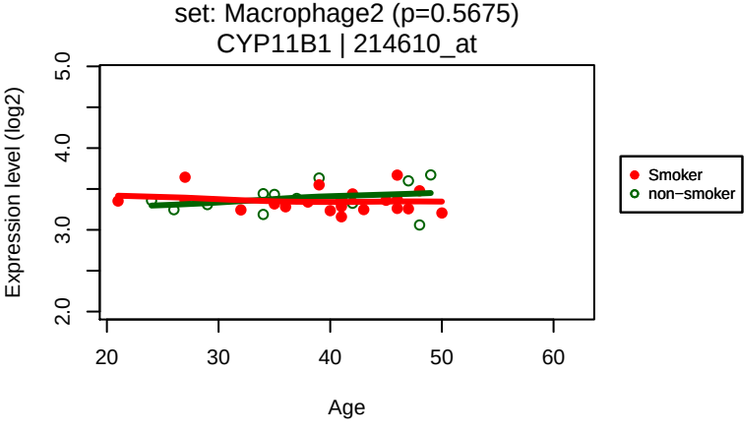

set: SmallAirway2 (p=0.3419)
CYP11B1 | 214610_at
Expression level (log2)
Age
Smoker
non−smoker

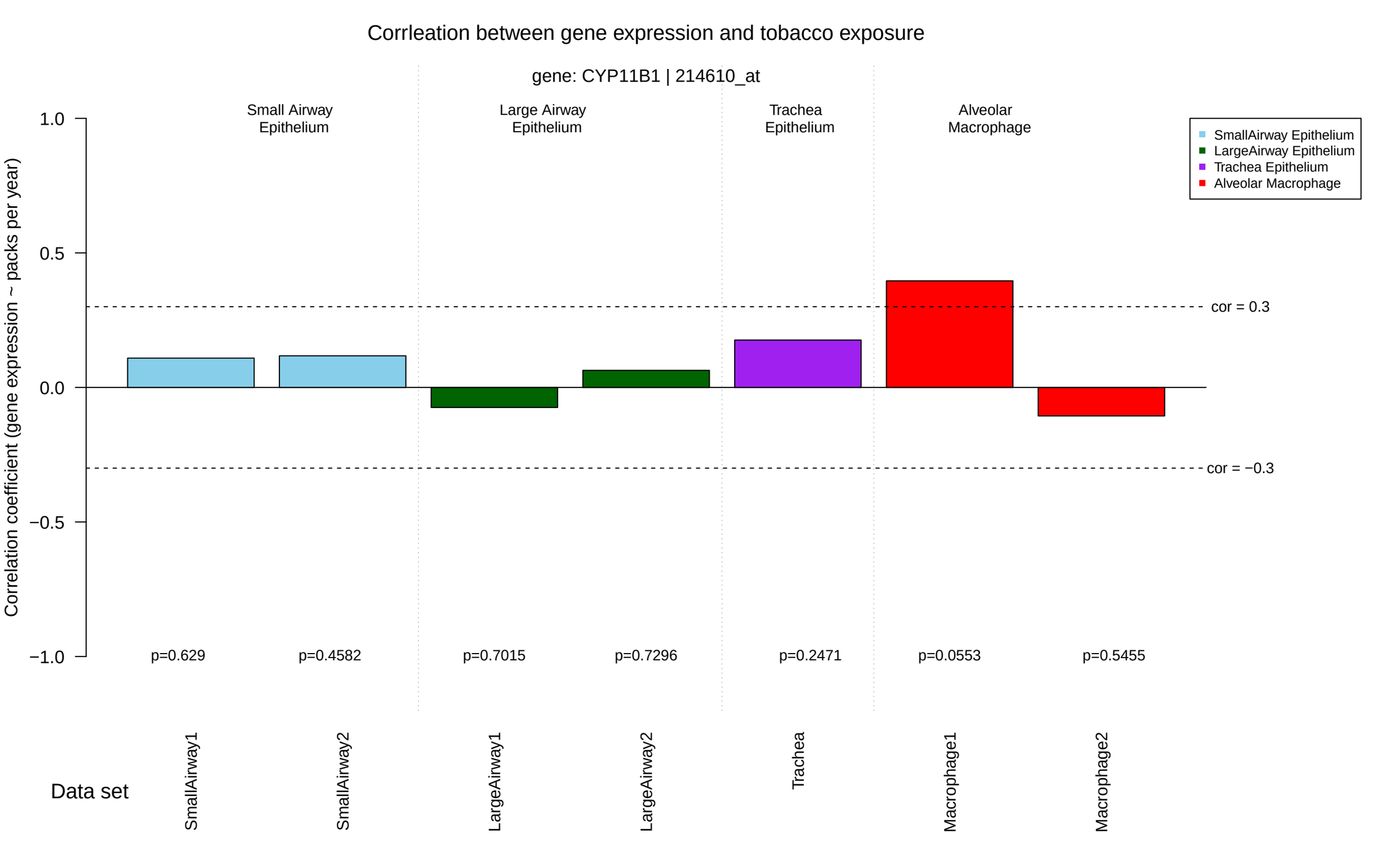

set: LargeAirway1 (p=0.2292)
CYP11B1 | 214610_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0936)
CYP11B1 | 214610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1493)
CYP11B1 | 214610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0621)
CYP11B1 | 214610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5675)
CYP11B1 | 214610_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP11B1 | 214610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.629
p=0.4582
p=0.7015
p=0.7296
p=0.2471
p=0.0553
p=0.5455
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2